Andrés Santiago Martínez Hernández

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Biomedical Engineer | AI & Automation Consultant | Digital Neuroscience Researcher

PROFESSIONAL & RESEARCH EXPERIENCE

CEO & Co-founder | *ART/Wear* — 2023 – Present

- Managing an international startup corporate structure alongside a cross-functional leadership team.
- Raised over 100K in grants and successfully developed patents, industrial designs, and proprietary software.
- Pitched the mechatronic smart garment platform at the VivaTech conference in Paris to pre-seed angel networks and accelerators.

Independent AI & Automation Consultant | *Freelance* — 2024 – Present

- Offering strategic consultancy to C-level teams in health and engineering firms to upgrade technical capacities.
- Building complex data architectures to ensure robust Electronic Data Processing (EDP) and systemic optimization through data mining.

Professional Researcher | *Universidad de los Andes, Faculty of Medicine* — 2024 – 2025

- Acted as a biomedical engineer researcher starting in 2024.
- Structured datasets and conducted epidemiological research, GPS, IMU, and wearables' signal analysis for the "On the Way to School" project.

Teaching Assistant & Prototypes Consultant | *Universidad de los Andes, Dept. of Biomedical Engineering* — 2025

- Served as an innovation advisor and teaching assistant for the "Introduction to Biomedical Engineering" course.
- Acted as a prototypes consultant on the design project coursework, guiding students in hardware development and execution.

Assistant to the Main Branch Representative | *Universidad Nacional de Colombia, Faculty of Economic Sciences* — 2024 – 2025

- Represented the main university branch for the Faculty of Economic Sciences to direct international relationships.
- Authored and managed cooperation and learning agreements with universities, research institutions, and corporate entities.

Max Planck International Fellow | *Max Planck Institute for Infection Biology, Berlin* — 2024

- Participated as an international fellow in Berlin, Germany.
- Engaged in a highly formative research experience ("eine prägende Erfahrung") contributing to advanced biological studies and international collaboration.

Research Intern | *Barrow Neurological Institute* — 2023

- Conducted research on precision Parkinson's disease prognosis.
- Applied both clinical and analytical skills in the field of movement disorders, specifically studying Parkinson's Disease and Essential Tremor.
- Assessed disease and pharmacological treatment effects using advanced structural and functional imaging biomarkers.

EDUCATION

Specialized Master's in Digital Neuroscience | *University of Fribourg, Switzerland* — 2026 – Present

Engineer's Degree, Computational and Software Engineering | *Universidad de los Andes, Colombia* — 2021 – 2024

- GPA: 4.05 / 5.0

Engineer's Degree, Biomedical Engineering (Minor in Neuroscience) | *Universidad de los Andes, Colombia* — 2019 – 2023

- GPA: 4.1 / 5.0

Selected Continuing Education & Fellowships:

- **Venturelab:** Completed Business Concept and Business Creation educational frameworks, Lausanne.
- **Neuromatch Impact Scholars Program:** Engineered a Google Colab Pro signal-processing pipeline downsampling high-resolution 16-channel sEMG array inputs for continuous 3D hand-pose estimation.
- **OxML:** Summer school at Oxford Machine Learning: Health track and cases, 2024.

SELECTED TEACHING EXPERIENCE

- **Neuromatch Academy:** Teaching Assistant for the "Computational Neuroscience" course for two consecutive years (including 2026).
- **Universidad de los Andes:** Principal teacher of "Computer Vision and Artificial Intelligence fundamentals," 2023.
- **Upcoming Autumn Seminar:** Accepted to present "The unfair advantage: Practical AI Automation" at the University of Fribourg Teach & Learn Together Days.

HONORS, GRANTS & AWARDS

- Innovation grant, Innovation club Fribourg (1K award)
- New Talents Research Award, WTW Insurance, 2026 (6K award).
- Startup Chile Innovation Grant, CORFO & Chilean Government, 2025.
- GSEA Entrepreneur of the Year (Colombia), Global Student Entrepreneurship Awards, 2024.
- First Place in Innovation, French Prix Bogotá, 2024.
- MPIIB Scholarship, Max Planck Institut für Infektionsbiologie, 2024.

TECHNICAL SKILLS & PROFICIENCIES

- **Programming & Development:** Expert in Microsoft Power Platform. Python (AI and data science), R-Project, Java, C, C++, SQL, HTML, CSS. Flutter, Web Audio API, Google Apps Script.
- **Tools & Platforms:** Jupyter Notebooks, GitHub, ROS-2, PostgreSQL, MATLAB.
- **Hardware & Engineering:** Mechatronics, Computer Vision, SDM manufacturing. Fusion 360, EAGLE, AutoCAD. Microelectronics prototyping, 3D printing, firmware flashing.

Languages: Native Spanish; English (Certified C2 by IELTS and TOEFL-IBT); Basic Italian.